



A Level • Cambridge (CIE) • Chemistry

38 mins 20 questions

Multiple Choice Practice

Topical Practice MCQ

15. Halogen compounds

Carefully selected past paper questions by topic.

- 28** What is involved in the mechanism of the reaction between aqueous NaOH and 1-bromobutane?
- A** attack by a nucleophile on a carbon atom with a partial positive charge
 - B** heterolytic bond fission and attack by a nucleophile on a carbocation
 - C** homolytic bond fission and attack by an electrophile on a carbanion
 - D** homolytic bond fission and attack by a nucleophile on a carbocation

31 The table shows three sets of reagents and reaction conditions.

	reagents	reaction conditions
1	$\text{CH}_2\text{C}(\text{CH}_3)\text{CH}_3$ and $\text{HCl}(\text{g})$	room temperature
2	$\text{CH}_3\text{C}(\text{CH}_3)(\text{OH})\text{CH}_3$ and SOCl_2	room temperature
3	$\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$ and Cl_2	the presence of ultraviolet light

Which sets of reagents and conditions can be used to produce 2-chloro-2-methylpropane as one of the organic products?

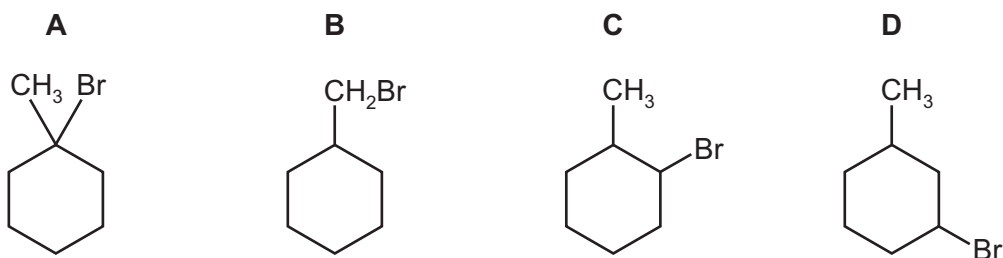
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

27 Compound X, $C_7H_{13}Br$, reacts with hot alcoholic NaOH to produce two compounds, Y and Z.

On reaction with Br_2 , Y gives a product, $C_7H_{12}Br_2$, which exists as a mixture of four optical isomers.

On reaction with Br_2 , Z gives a product, $C_7H_{12}Br_2$, which is non-chiral.

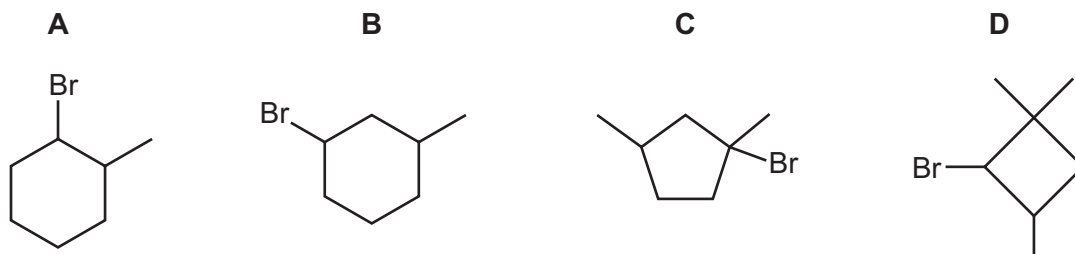
What could X be?



- 28 Compound Z, $C_7H_{13}Br$, has two chiral centres. A sample of Z contains all four possible optical isomers.

This sample of Z reacts with hot ethanolic NaOH to produce a mixture of **only** three isomers. Two of these isomers are optical isomers of each other.

What could be the formula of Z?



20 Bromoethane reacts with cyanide ions, producing propanenitrile.

Which statement about the S_N2 mechanism of this reaction is correct?

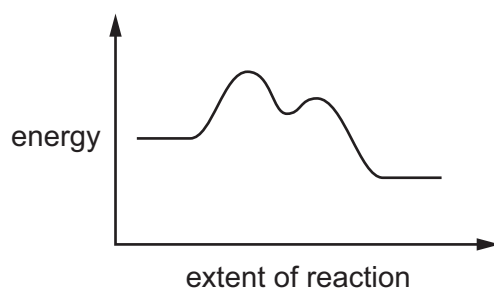
- A** The lone pair of electrons on C of CN^- attacks the carbon atom of the C–Br bond.
- B** The lone pair of electrons on C of CN^- attacks the carbocation formed when the C–Br bond breaks.
- C** The lone pair of electrons on N of CN^- attacks the carbon atom of the C–Br bond.
- D** The lone pair of electrons on N of CN^- attacks the carbocation formed when the C–Br bond breaks.

24 Bromoethane and chloroethane are added separately to water. Hydrolysis reactions occur.

Which compound hydrolyses more rapidly and what is the mechanism?

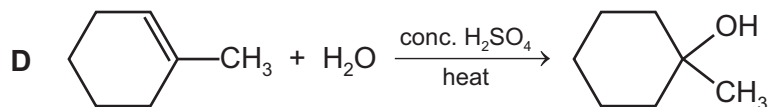
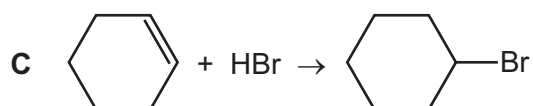
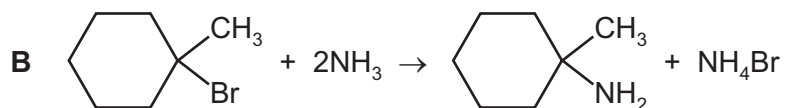
	compound that hydrolyses more rapidly	mechanism
A	bromoethane	electrophilic substitution
B	bromoethane	nucleophilic substitution
C	chloroethane	electrophilic substitution
D	chloroethane	nucleophilic substitution

25 A reaction pathway diagram is shown.

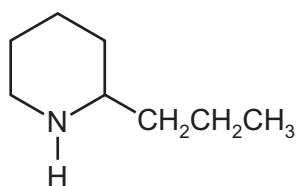


The four reactions that follow are all exothermic.

Which reaction would **not** have such a pathway?

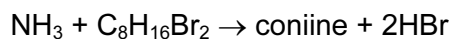


24 Coniine is the major constituent of the poison 'oil of hemlock'.



coniine

Coniine can be synthesised by reacting ammonia with a dibromo compound, **X**.

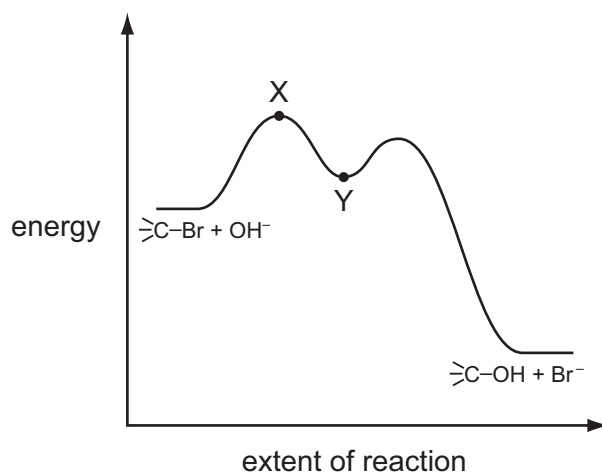


X

What is the name of compound **X**?

- A** 1,1-dibromo-2-propylcyclopentane
- B** 1,2-dibromo-2-propylcyclopentane
- C** 1,4-dibromooctane
- D** 1,5-dibromooctane

- 28 A tertiary bromoalkane, indicated here by >C-Br , reacts with aqueous NaOH. The mechanism has the reaction pathway below.



Which point in the diagram is correctly identified?

A X is >C^+

B X is $\left[\text{HO} \cdots \text{C} \cdots \text{Br} \right]^-$

C Y is >C^+

D Y is $\left[\text{HO} \cdots \text{C} \cdots \text{Br} \right]^-$

- 25 In the hydrolysis of bromoethane by aqueous sodium hydroxide, what is the nature of the attacking group and of the leaving group?

	attacking group	leaving group
A	electrophile	electrophile
B	electrophile	nucleophile
C	nucleophile	electrophile
D	nucleophile	nucleophile

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B	electrophile	nucleophile
C	nucleophile	electrophile
D	nucleophile	nucleophile

21 Which reaction will give the best yield of 1-chloropropane?

- A** chlorine gas with propene gas in the dark
- B** propan-1-ol with dilute $\text{NaCl}(\text{aq})$
- C** propan-1-ol with PCl_5
- D** propene with dilute $\text{HCl}(\text{aq})$

24 Including structural and stereoisomers, how many isomeric products are produced when alcoholic KOH reacts with 2-chlorobutane?

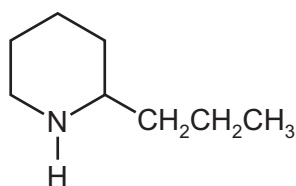
A 1

B 2

C 3

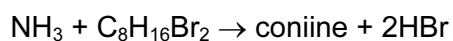
D 4

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X

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- C** 1,4-dibromooctane
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22 Which reaction will give 2-chloropropane in the best yield?

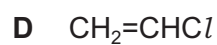
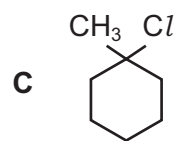
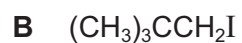
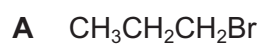
- A** propane gas with chlorine gas in the presence of ultraviolet light
- B** propan-2-ol with dilute $\text{NaCl}(\text{aq})$
- C** propan-2-ol with SOCl_2
- D** propene with dilute $\text{HCl}(\text{aq})$

- 29 When an isomer Y of molecular formula $\text{C}_4\text{H}_9\text{Br}$ undergoes hydrolysis in aqueous alkali to form an alcohol $\text{C}_4\text{H}_9\text{OH}$, the rate of reaction is found to be unaffected by changes in the concentration of OH^- ions present.

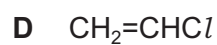
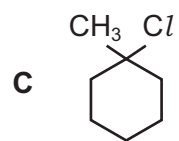
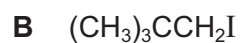
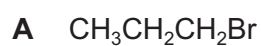
Which is the most likely molecular structure of Y?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$
- B $\text{CH}_3\text{CH}_2\text{CHBrCH}_3$
- C $(\text{CH}_3)_2\text{CHCH}_2\text{Br}$
- D $(\text{CH}_3)_3\text{CBr}$

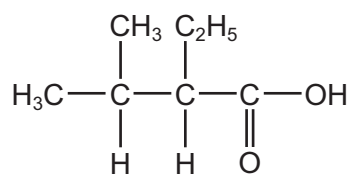
28 Which compound undergoes an S_N1 substitution reaction?



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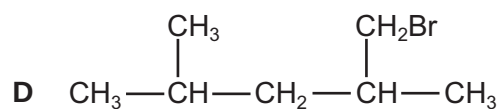
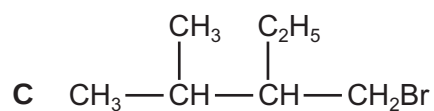
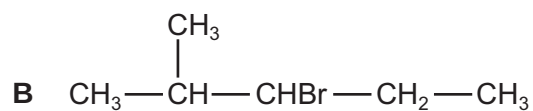
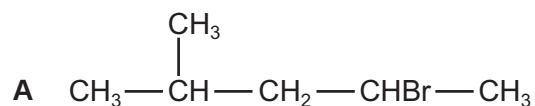


- 29 The characteristic odour of rum is attributed to the compound 2-ethyl-3-methylbutanoic acid.

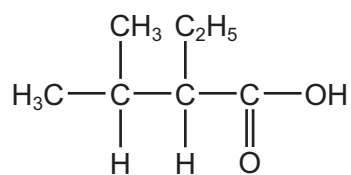


2-ethyl-3-methylbutanoic acid

Which compound will produce 2-ethyl-3-methylbutanoic acid by heating under reflux with alcoholic sodium cyanide and subsequent acid hydrolysis of the reaction product?

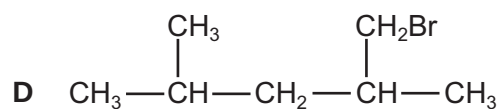
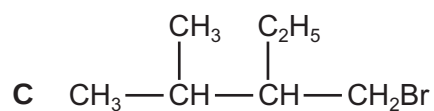
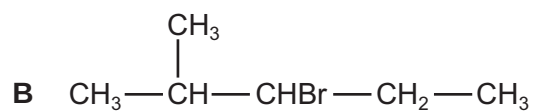
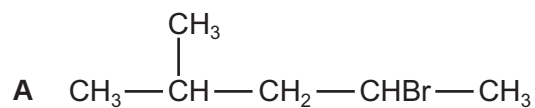


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2-ethyl-3-methylbutanoic acid

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Answer Key

No.	Paper	Topic	Answer
1	9701_s24_qp_13	15. Halogen compounds	A
2	9701_w24_qp_12	15. Halogen compounds	A
3	9701_m23_qp_12	15. Halogen compounds	A
4	9701_s22_qp_12	15. Halogen compounds	A
5	9701_s21_qp_11	15. Halogen compounds	A
6	9701_w19_qp_11	15. Halogen compounds	B
7	9701_s16_qp_13	15. Halogen compounds	A
8	9701_s15_qp_11	15. Halogen compounds	D
9	9701_s15_qp_13	15. Halogen compounds	C
10	9701_w14_qp_11	15. Halogen compounds	D
11	9701_w14_qp_12	15. Halogen compounds	D
12	9701_w13_qp_11	15. Halogen compounds	C
13	9701_w13_qp_13	15. Halogen compounds	C
14	9701_s12_qp_12	15. Halogen compounds	D
15	9701_s11_qp_12	15. Halogen compounds	C
16	9701_s10_qp_13	15. Halogen compounds	D
17	9701_w10_qp_11	15. Halogen compounds	C
18	9701_w10_qp_13	15. Halogen compounds	C
19	9701_w09_qp_11	15. Halogen compounds	B
20	9701_w09_qp_12	15. Halogen compounds	B